Chemistry of the Earth: Lecture 10

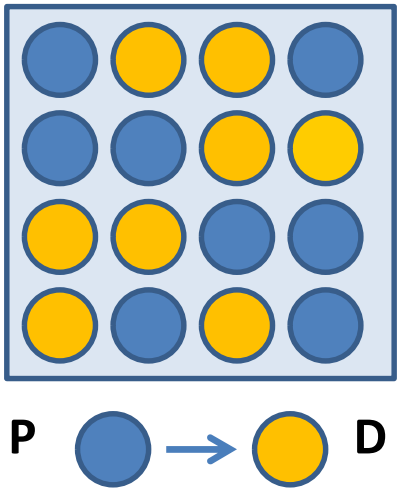
Radioactive isotopes and Geochronology



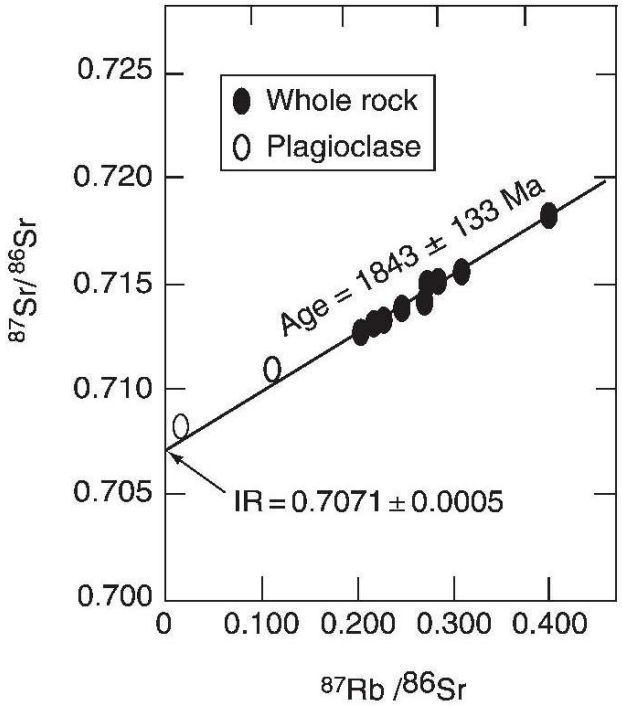
Part 1: Radioactive decay

OBJECTIVES OF THIS LECTURE AND PRACTICAL

- Consider what the "age" of a rock means
- Review how radioactive isotopes decay
- Understand how radioactive isotopes are used to determine the age of a rock – geochronometry
- Investigate which elements and isotopes make useful geochronometers, and why



$$D = D_0 + N (e^{\lambda t} - 1)$$



READING AND RESOURCES

Grotzinger & Jordan	**Chapter 9, p. 253-258**

(Recommended, basic reading)

Misra	**Chapter 10.1-10.3**
McSween & Richardosn	**Chapter 14, p. 286-296**

(Both recommended, more detailed)

Gill	Chapter 10, p. 179-189
White	Chapter 8

Further Reading: Elements 2013, One Hundred Years of Geochronology.
http://www.elementsmagazine.org/archivearticles/e9_1/e9_1.pdf

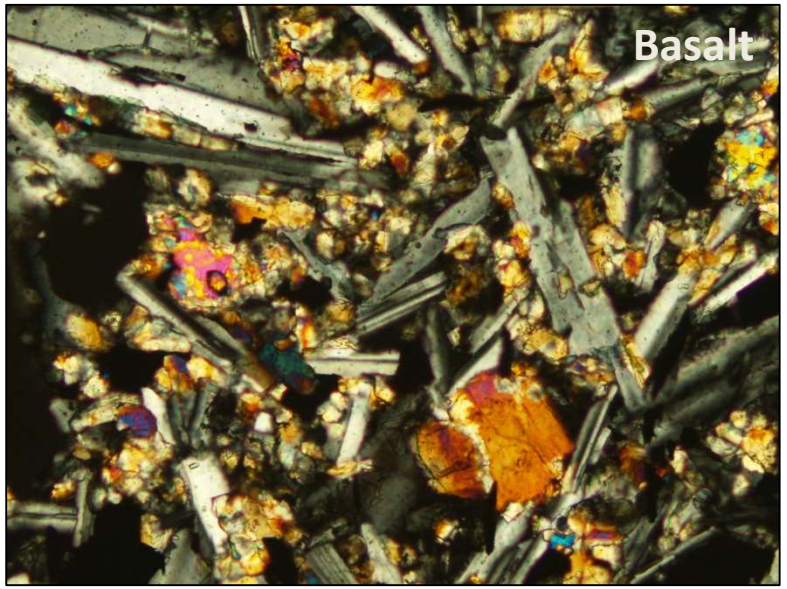
What do we mean when we talk about the "age" of a rock?

Igneous rocks

The "age" of an igneous rock refers to the time when it solidified from a molten state. This is the time when the minerals in the rock formed.



http://hvo.wr.usgs.gov/archive/2004_07_28.html



Crossed polarizers: 1mm across

Extrusive Igneous Rock: e.g. Basalt

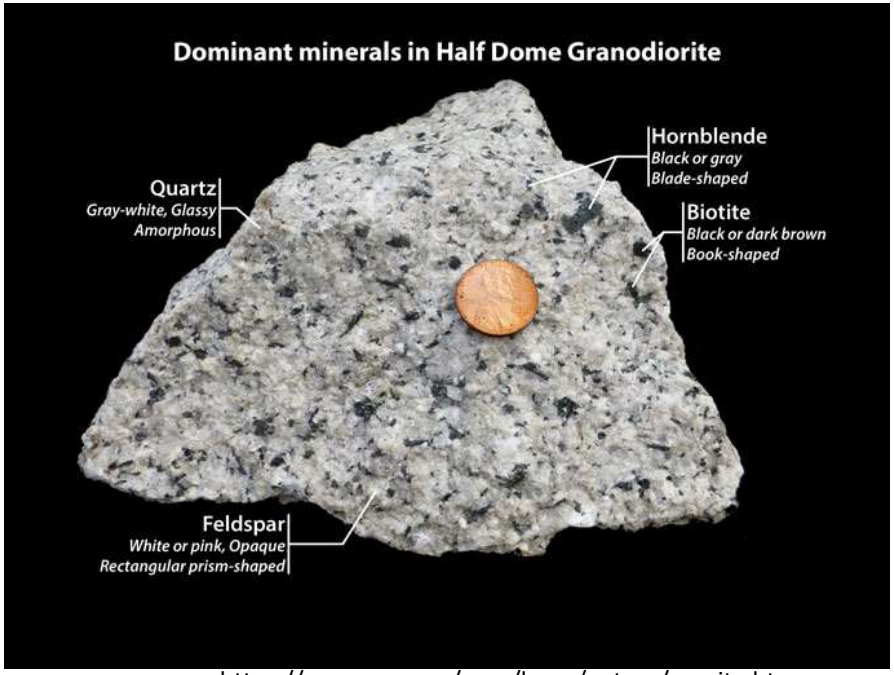
- Rock is 10 years old
- Age is 10 years
- Rock formed 10 years ago

Intrusive Igneous Rock: e.g. Granite

Age is the time when the mineral grains grew as the rock cooled



https://www.flickr.com / Johan V



https://www.nps.gov/yose/learn/nature/granite.htm

For example:

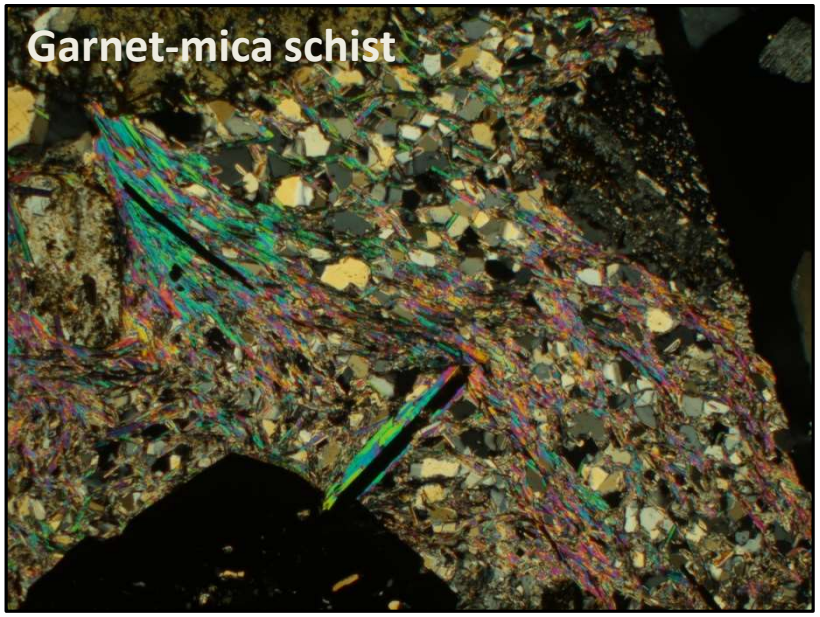
- Rock is 100 million years (100 M.y.) old
- Age is 100 million years (100 Ma or 0.1 Ga)
- Rock formed 100 million years ago

Metamorphic rocks

The “age” of a metamorphic rock refers to the time
when the *minerals that are being dated* formed.
Metamorphic rocks can retain a record of continuous change in pressure, temperature and fluid composition. Different minerals in the rock might have formed at different times.



http://www.discovergrandteton.org/teton-geology/teton-features/



Crossed polarizers: 2mm across

For example:

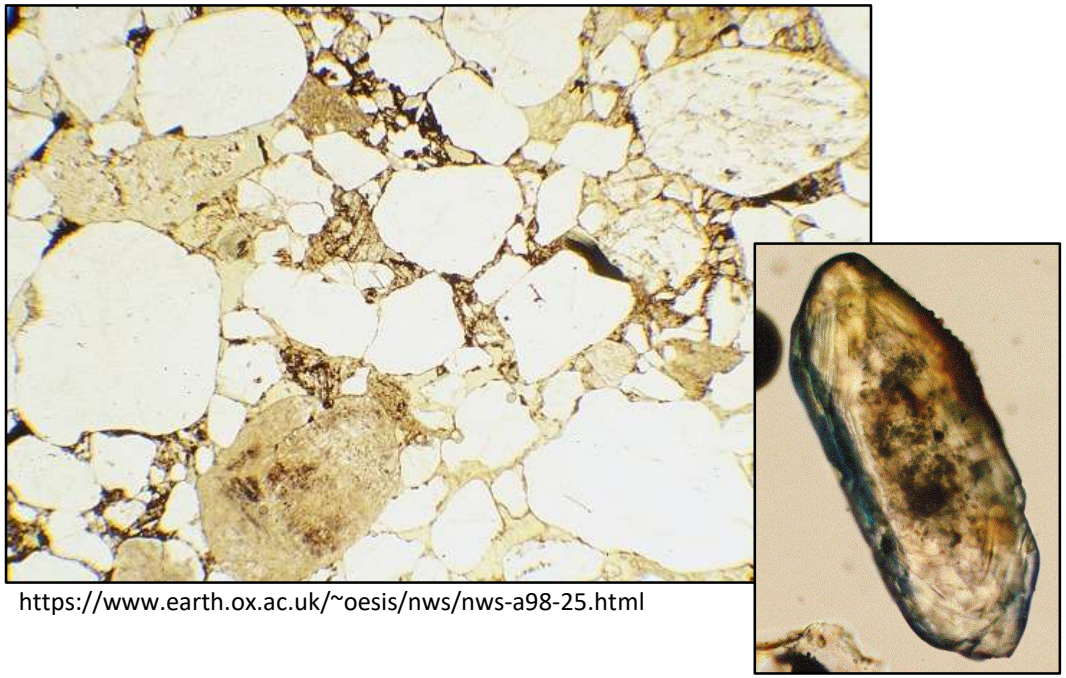
- Rock is 2,500 million years (2,500 M.y.) old
- Age is 2,500 million years or 2.5 billion years (2,500 Ma or 2.5 Ga)
- Rock formed 2,500 million years ago

Sedimentary rocks

The meaning of an “age” for a sedimentary rock needs to be specified. Sedimentary rocks consist of grains from a range of different source rocks that each has its own age. The rock also has a cement that obviously was the last material to form.



Photograph copyright Bruce Watson, Rensselaer



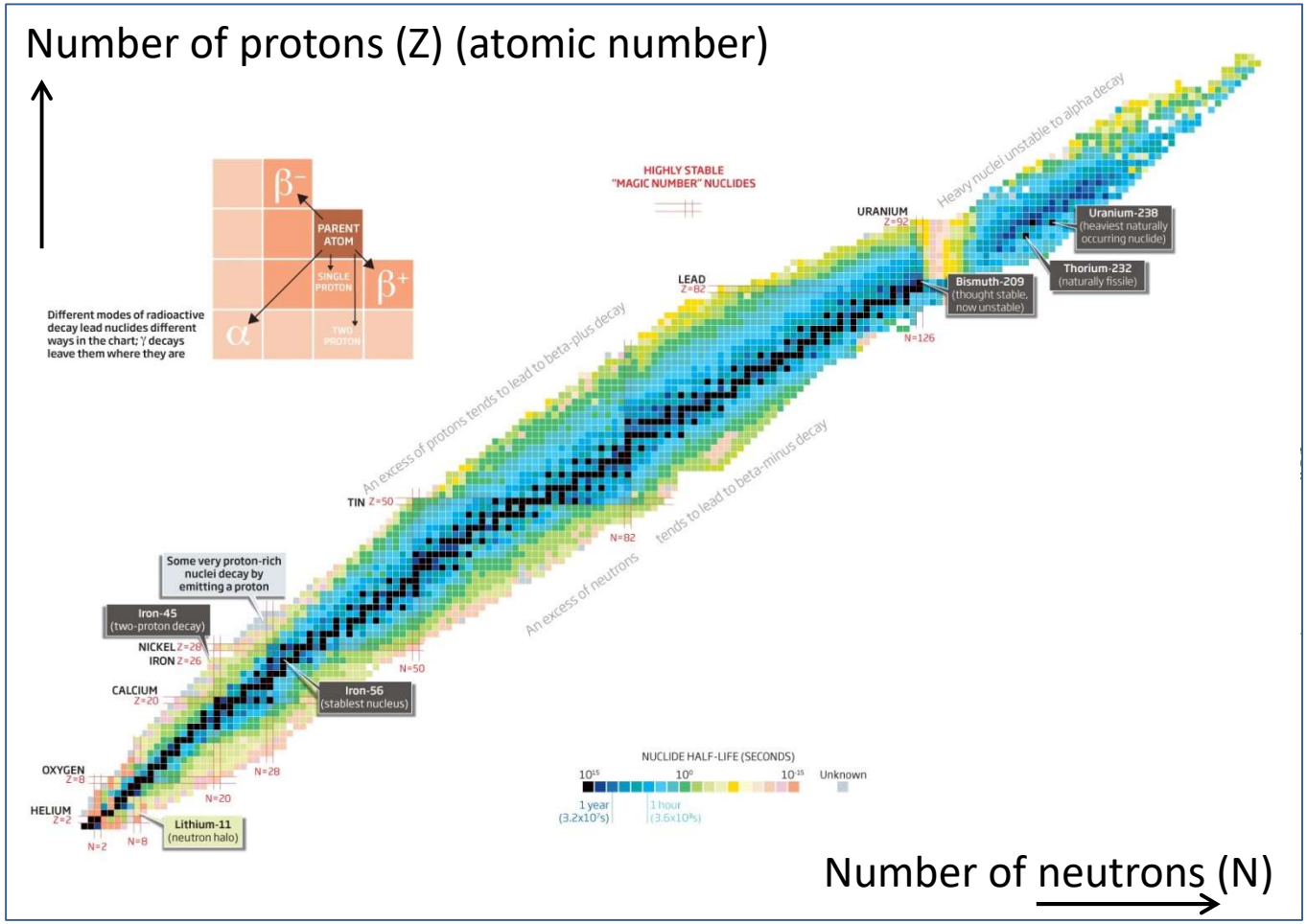
https://www.earth.ox.ac.uk/~oesis/nws/nws-a98-25.html

Zircon grain, 250 μm long.
Wikipedia

“The bits and pieces of rocks that make up the Jack Hills rock formation are ancient—over 3 billion years old. Individual crystals of zircon within the rocks are 4.4 billion years old, only 150 million years or so younger than the age of the Earth itself. These crystals are the oldest fragments of the Earth yet found.”

http://earthobservatory.nasa.gov/Features/Zircon/

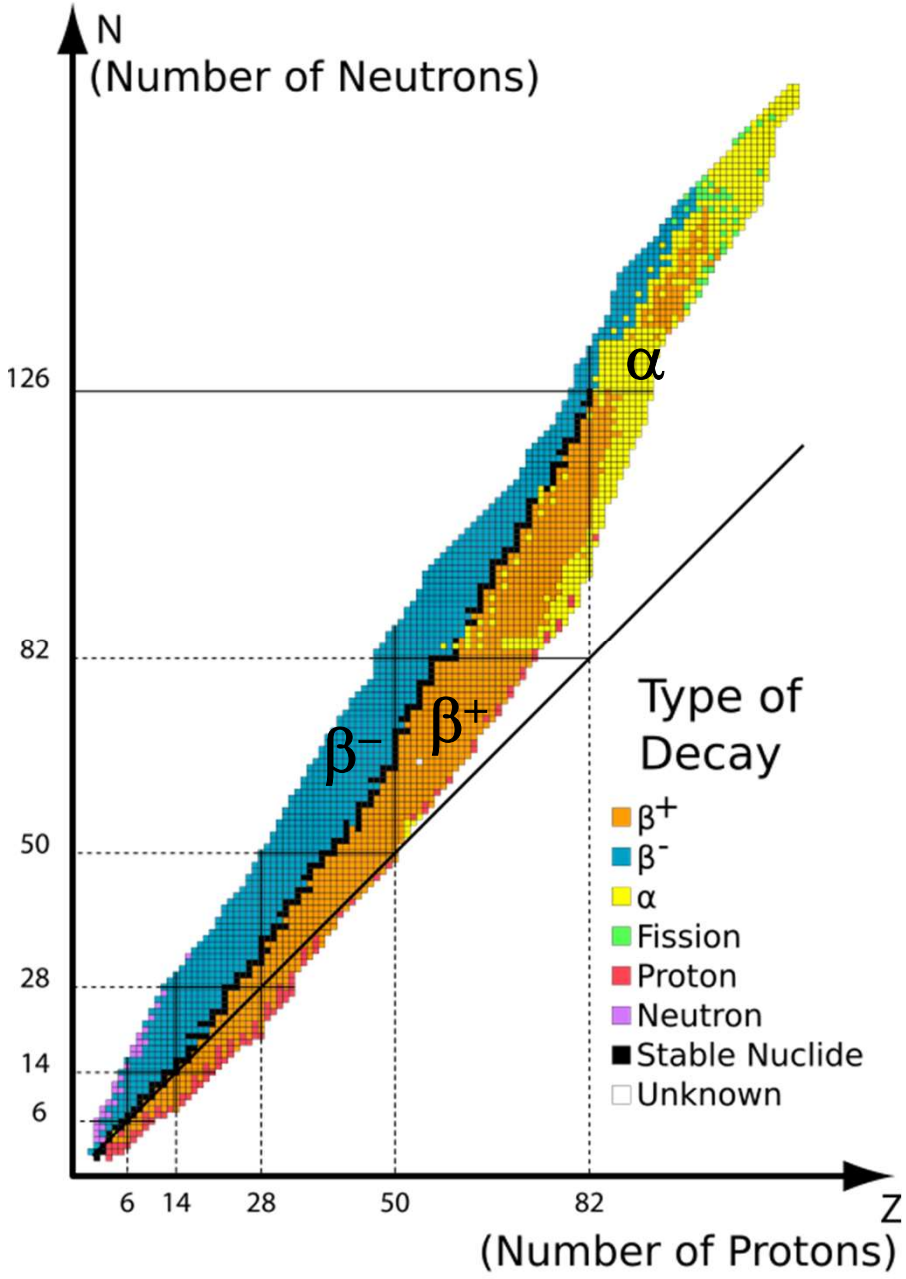
Radioactive isotopes:
Isotopes that undergo natural radioactive decay



http://www.nuclear-power.net

- Chart shows known isotopes. Black squares are stable isotopes.
- Coloured squares are radioactive isotopes. These will decay naturally, until stable isotopes are produced.

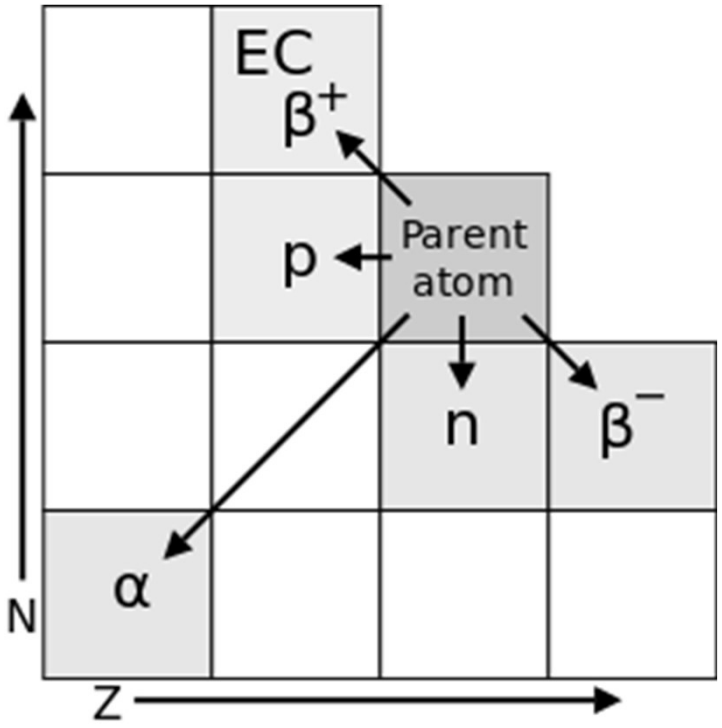
- Parent isotope decays to daughter isotope.
- Radioactive isotope (radionuclide) decays to radiogenic isotope.



Note: This chart is flipped compared with the previous one (N vs Z not Z vs N)

The most common types of decay are alpha (α) and beta (β) decay.

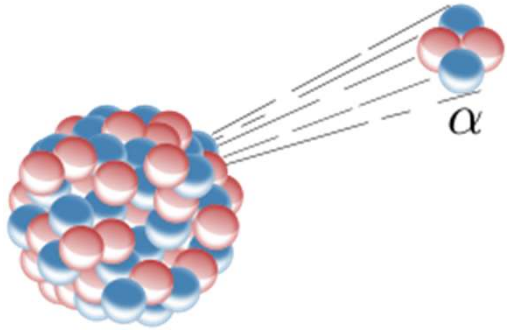
Different modes of radioactive decay lead different ways in the chart. Decay pathways lead to stable nuclides.



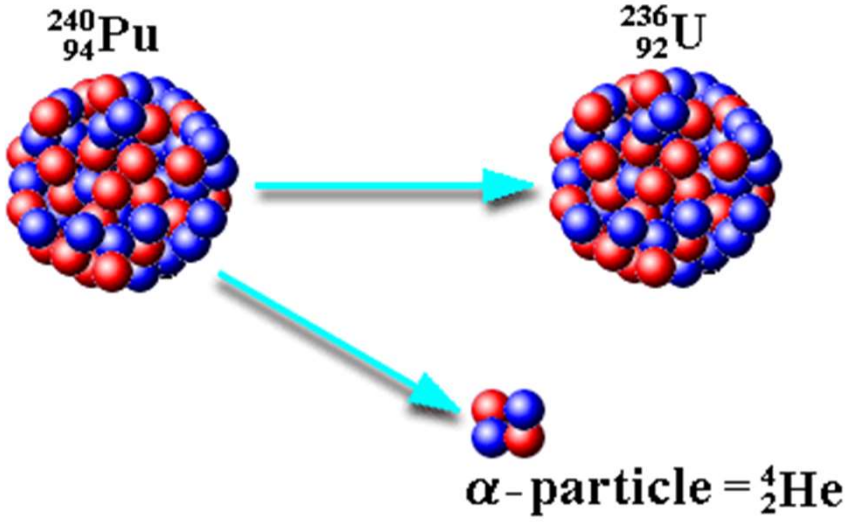
Alpha decay

A nucleus loses an alpha particle (α).

Alpha particles are equivalent to
He nuclei: 2 protons + 2 neutrons



Wikipedia



Example: ^{240}Pu decays to ^{236}U.

Loss of an α particle means that:

- the atomic mass (A) decreases by 4
- the atomic number (Z) decreases by 2.

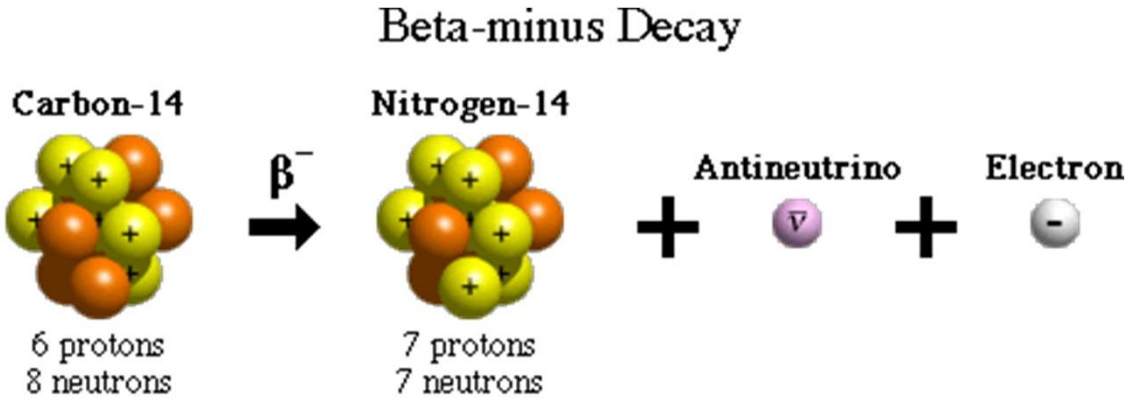
http://www.nuceng.ca/igna/radioactivity.htm

Beta minus decay

A neutron in the nucleus loses a beta-minus (β^-) particle, which is equivalent to an electron. The neutron turns into a proton.

$$n \rightarrow p^+ + \beta^- + \overline{\nu}$$

neutron proton electron antineutrino



Example: ^{14}C decays to ^{14}N.

Loss of a β^- particle means that:

- the atomic mass (A) stays the same
- the atomic number (Z) increases by 1

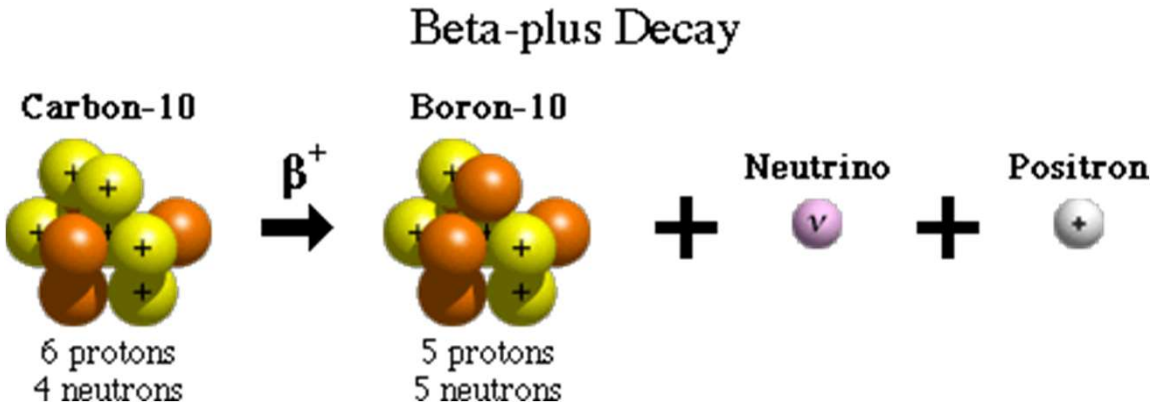
http://education.jlab.org/glossary/betadecay.html

Beta plus decay

A proton in the nucleus loses a beta-plus (β^+) particle, which is a positron. The proton turns into a neutron.

$$p^+ \rightarrow n + \beta^+ + \nu$$

proton neutron positron neutrino



Example: ^{10}C decays to ^{10}B.

Loss of a β^+ particle means that:

- the atomic mass (A) stays the same
- the atomic number (Z) decreases by 1.

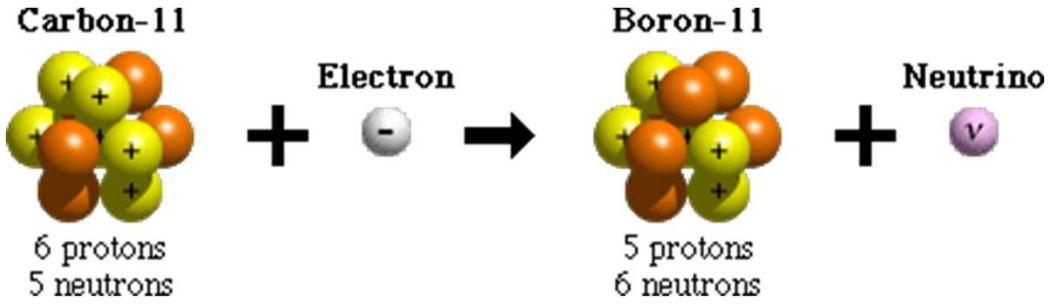
http://education.jlab.org/glossary/betadecay.html

Electron capture

Capture of an electron by the nucleus. A proton in the nucleus is converted into a neutron plus a neutrino.

$$p^+ + e^- \rightarrow n + \nu + \gamma$$

proton electron neutron neutrino gamma



Example: ^{11}C decays to ^{11}B.

Electron capture means that:

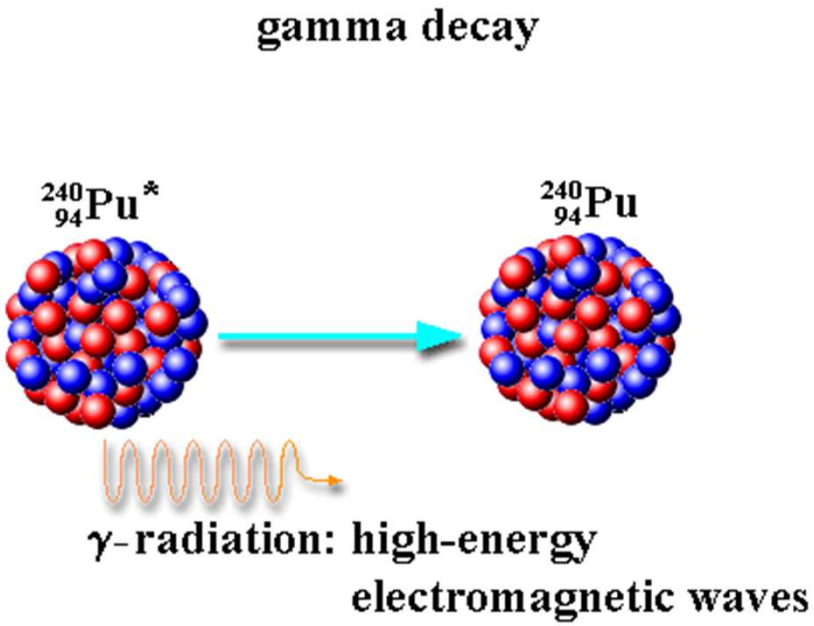
- the atomic mass (A) stays the same
- the atomic number (Z) decreases by 1.

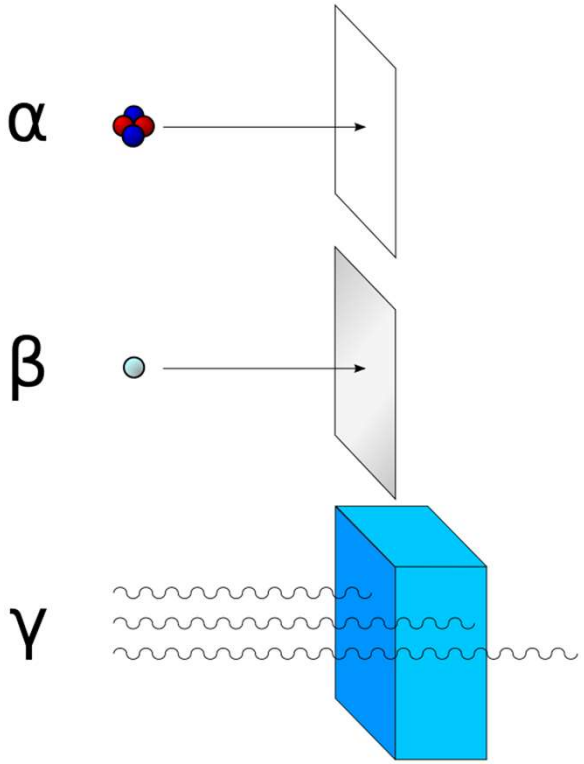
(This is the same net reaction as β^+ decay.)

http://education.jlab.org/glossary/electroncapture.html

Gamma decay

γ rays are not particulate radiation like α and β, but a form of high-energy electromagnetic wave.
γ decay results in no change in mass and no change in the isotope.





α particles are stopped by paper,
β particles by Al shielding,
γ rays by thick lead shielding.

http://www.nuceng.ca/igna/radioactivity.htm

Use of radioactive isotopes as geochronometers

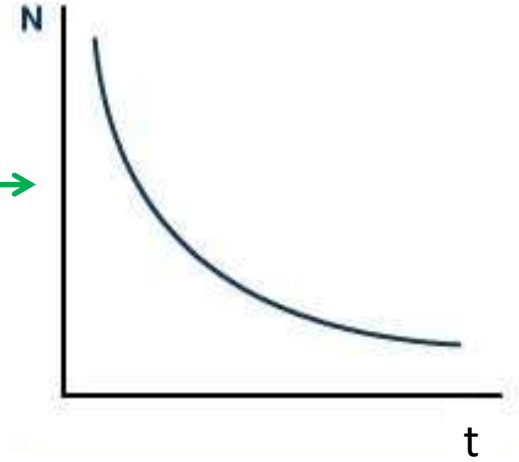
Radioactive decay:
The rate of decay of a radioactive nuclide at any instant is proportional to the number of atoms of that nuclide remaining at that instant.

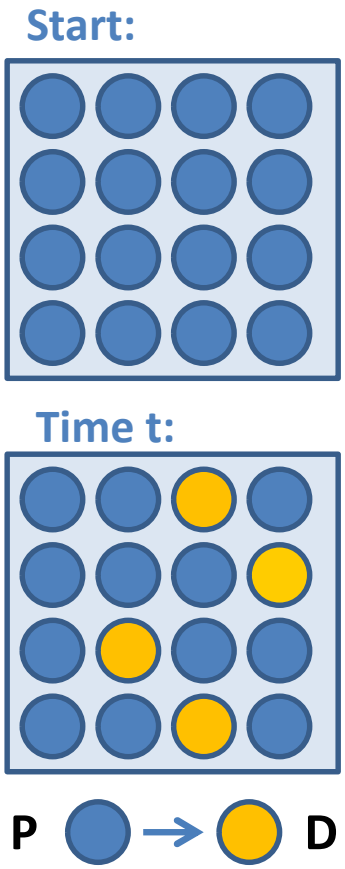
At time t, the number of parent atoms remaining is N.

$$-\frac{dN}{dt} \propto N \qquad \text{or} \qquad -\frac{dN}{dt} = \lambda N$$

λ is the constant of proportionality, called the **decay constant**

Graph showing exponential decay.
- The minus sign in the equation is because N is decreasing with time.
- dN/dt is the slope of the curve, which decreases with time and with the value of N.





Basic equation for radioactive decay:

1. Start with:

$$-\frac{dN}{dt} = \lambda N$$

2. Rearrange:

$$-\frac{dN}{N} = \lambda\, dt$$

3. Integrate this equation:

$$-\int dN / N = \lambda \int dt$$

$$-\ln N = \lambda t + C$$ (C is constant of integration) (Equation 1)

4. At time t = 0 (i.e. when decay begins), $N = N_0$. So, $\ln N_0 = C$

5. Substitute $\ln N_0 = C$ into equation (1) above

$$-\ln N = \lambda t + \ln N_0 \quad \text{or} \quad -\ln (N/N_0) = \lambda t$$

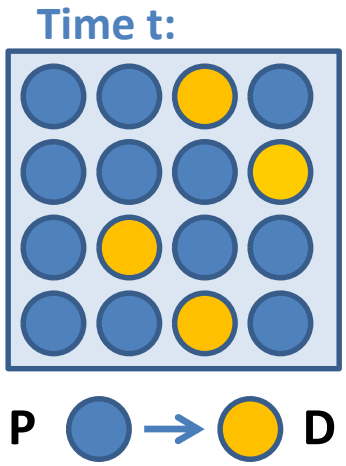
or

$$\boxed{N / N_0 = e^{-\lambda t}}$$

$$N / N_0 = e^{-\lambda t}$$

The equation relates the number of radioactive (parent) atoms (N), at any given time t, to the starting number of atoms (N_0 at t = 0) and the decay constant, λ.

Now we will consider the number of atoms of the daughter nuclide, D*.
(The * symbol is used to indicate radiogenic daughter atoms, i.e. ones produced by radioactive decay.)



At any given time t, $D^* = N_0 - N$

If we rearrange the main equation ,we have:

$$N_0 = N / e^{-\lambda t} \quad \text{or} \quad N_0 = N e^{\lambda t}$$

Substitute this into equation above, and we have:

$$D^* = N e^{\lambda t} - N \quad \text{or} \quad D^* = N (e^{\lambda t} - 1)$$

From the previous slide, $D^* = N(e^{\lambda t} - 1)$

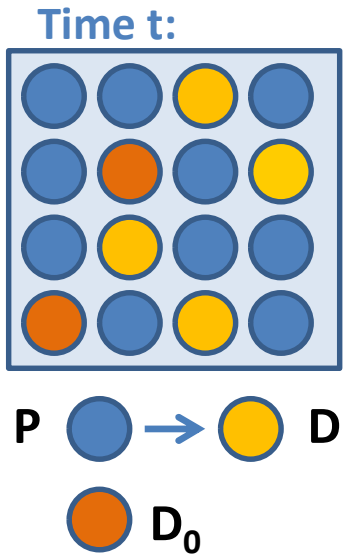
The equation relates the number of radiogenic daughter atoms (D*) to the number of parent atoms (N), at any given time t, and the decay constant, λ.

What if there were already some atoms of the daughter isotope present to begin with? We will call this amount D_0.
The total number of daughter atoms (D) present at time t is then given by:

$$D = D_0 + D^*$$

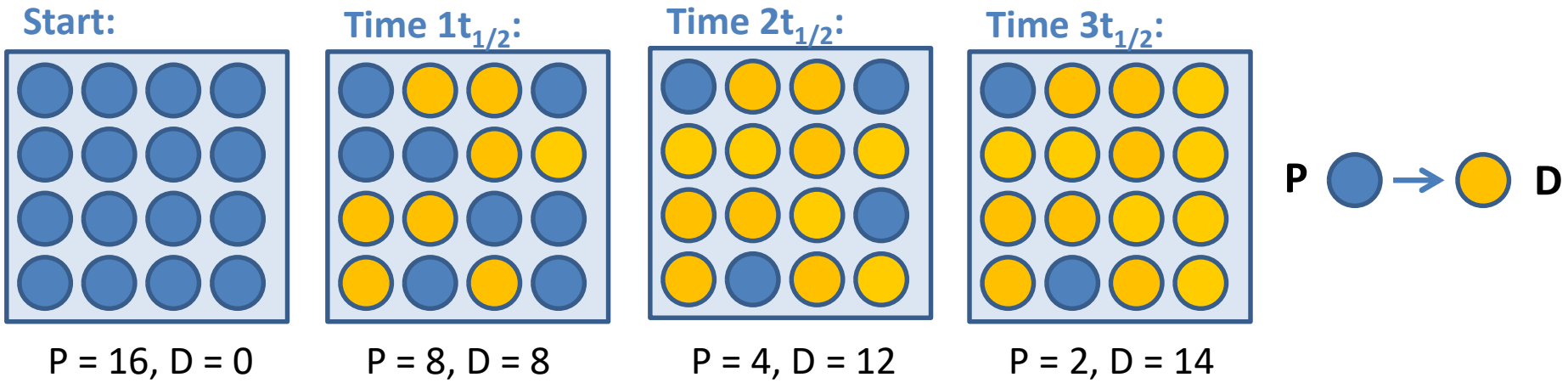
Combining this equation with the one at the top of the slide gives:

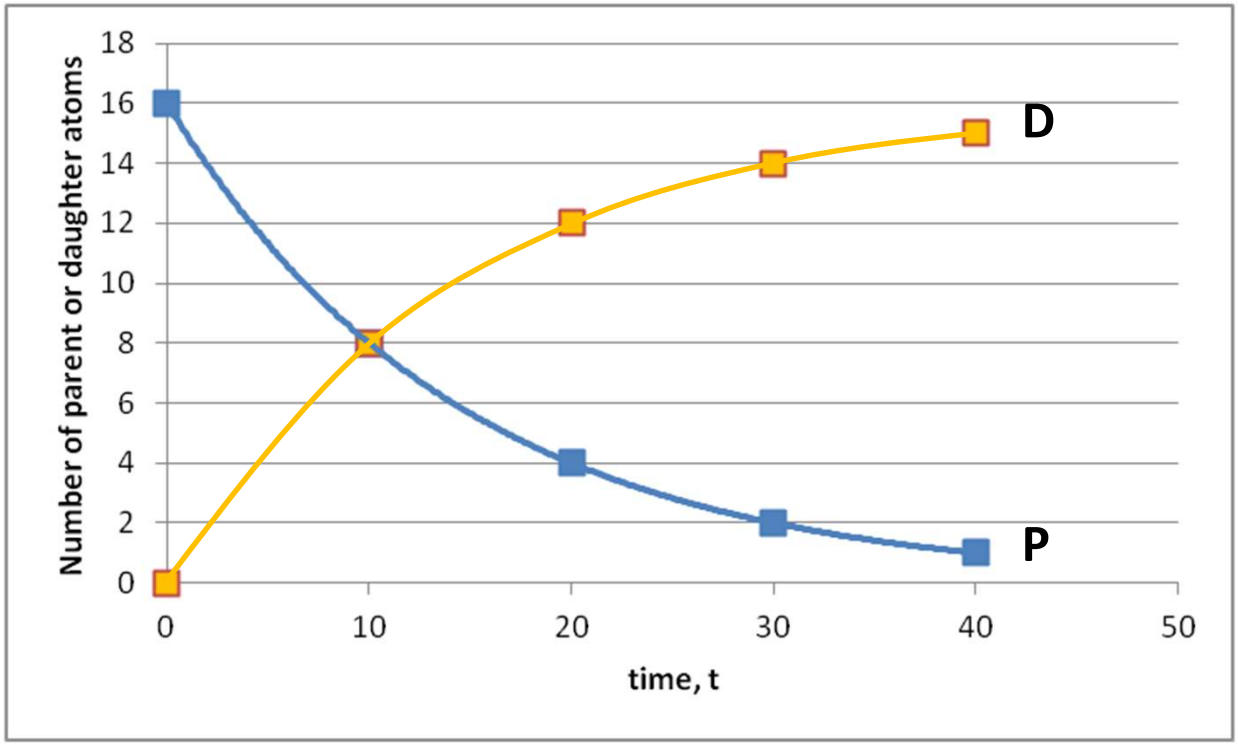
$$D = D_0 + N(e^{\lambda t} - 1)$$



This equation is the basis for **geochronology**.
We can measure the amounts of D and N in a rock.
D_0 is a constant and we can determine its value.

Half-life of radioactive decay:
The time it takes for half the radioactive atoms to decay





Relating the decay constant, λ, to the half-life of radioactive decay:

The half-life, $t_{½}$, is defined as the time required for half of the parent atoms to decay.

When $t = t_{½}$, $N = N_0 / 2$

Substituting into the equation $N / N_0 = e^{-\lambda t}$

gives $½ N_0 / N_0 = e^{-\lambda t ½}$

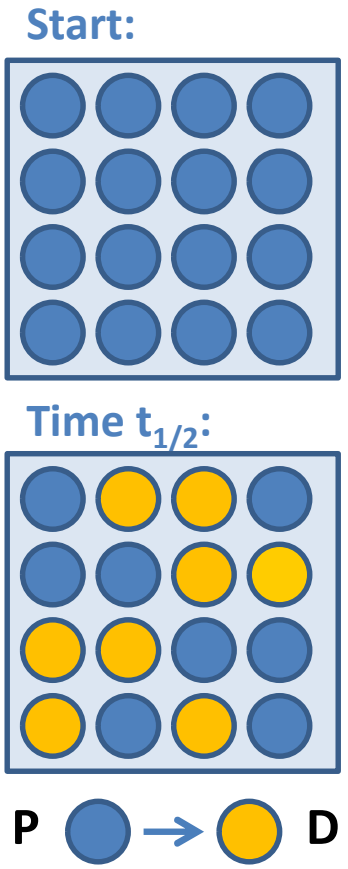
or $½ = e^{-\lambda t ½}$

Taking the natural log:

$\ln ½ = - \lambda t_{½}$ or $\ln 2 = \lambda t_{½}$

$t_{½} = \ln 2 / \lambda$ or $t_{½} = 0.693 / \lambda$

This equation gives the relationship between the **half-life** and the **decay constant**.



Chemistry of the Earth: Lecture 10

Radioactive isotopes and Geochronology



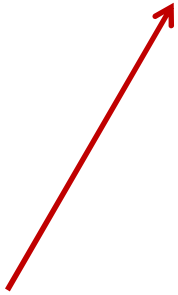
Part 2: Geochronology

Important geochronometers

Table 8.2 Long-lived radioactive decay systems of geochemical interest.

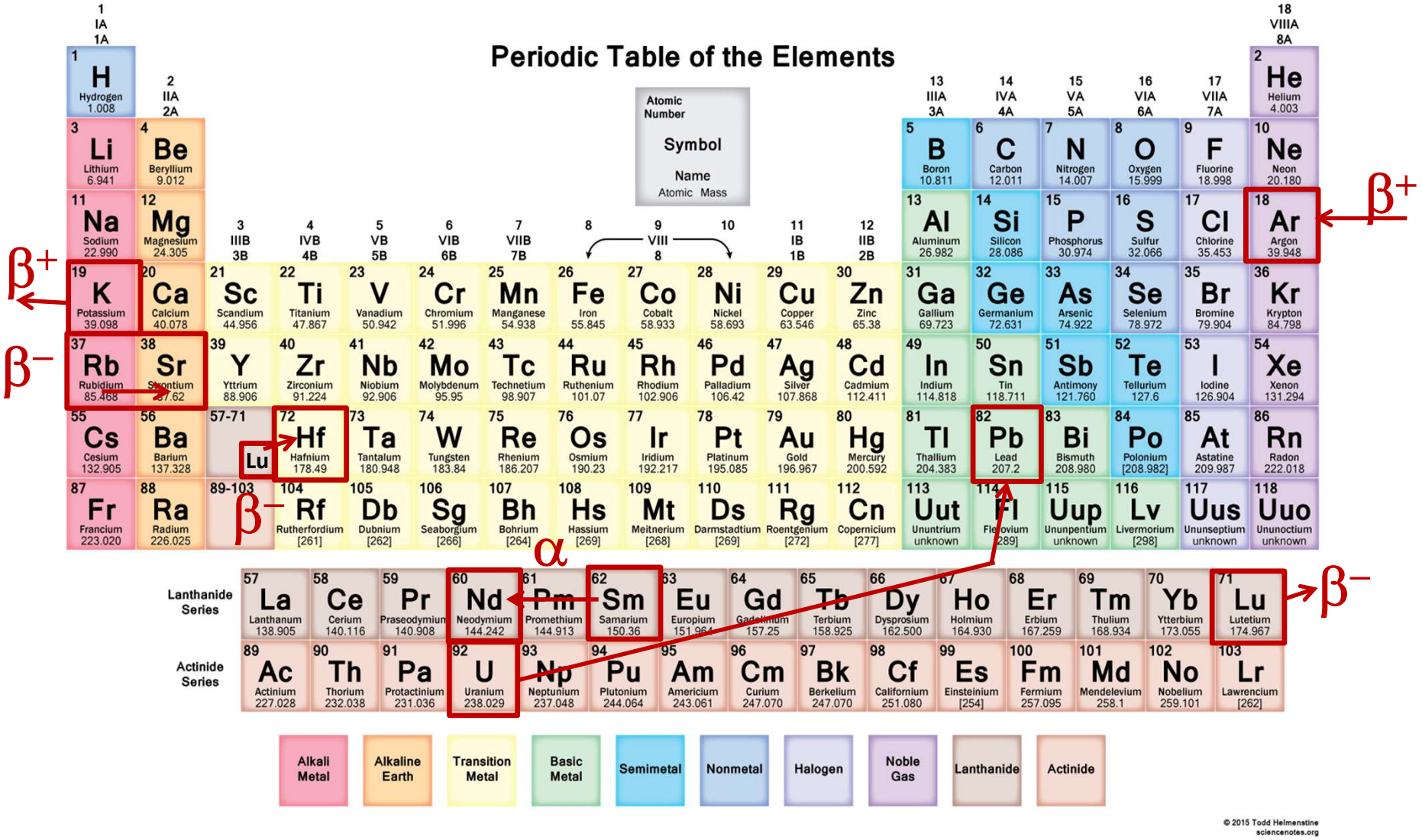
Parent	Decay mode	λ	Half-life	Daughter	Ratio
^{40}K	β^+, e.c., β^-	$5.5492 \times 10^{-10}\,y^{-1*}$	$1.25 \times 10^{9}\,y$	^{40}Ar, ^{40}Ca	$^{40}Ar/^{36}Ar$
^{87}Rb	β^-	$1.42 \times 10^{-11}\,y^{-1}$	$4.88 \times 10^{10}\,y$	^{87}Sr	$^{87}Sr/^{86}Sr$
^{138}La	β^-	$2.67 \times 10^{-12}\,y^{-1}$	$2.60 \times 10^{11}\,y$	^{138}Ce	$^{138}Ce/^{142}Ce$, $^{138}Ce/^{136}Ce$
^{147}Sm	α	$6.54 \times 10^{-12}\,y^{-1}$	$1.06 \times 10^{11}\,y$	^{143}Nd	$^{143}Nd/^{144}Nd$
^{176}Lu	β^-	$1.867 \times 10^{-11}\,y^{-1\ddagger}$	$3.71 \times 10^{10}\,y$	^{176}Hf	$^{176}Hf/^{177}Hf$
^{187}Re	β^-	$1.64 \times 10^{-11}\,y^{-1}$	$4.23 \times 10^{10}\,y$	^{187}Os	$^{187}Os/^{186}Os$
^{232}Th	α	$4.948 \times 10^{-11}\,y^{-1}$	$1.40 \times 10^{10}\,y$	^{208}Pb, ^{4}He	$^{208}Pb/^{204}Pb$, $^{3}He/^{4}He$
^{235}U	α	$9.849 \times 10^{-10}\,y^{-1\dagger}$	$7.04 \times 10^{8}\,y$	^{207}Pb, ^{4}He	$^{207}Pb/^{204}Pb$, $^{3}He/^{4}He$
^{238}U	α	$1.55125 \times 10^{-10}\,y^{-1}$	$4.47 \times 10^{9}\,y$	^{206}Pb, ^{4}He	$^{206}Pb/^{204}Pb$, $^{3}He/^{4}He$

Table from White



Ratio measured in the lab:
Daughter radiogenic isotope / Stable isotope of same element

Important radioisotopes used for geochronometry



Factors that make a good chronometer

Half-life

• Geochronometers are useful over a timescale up to about 10 half-lives

• If the half-life is very long we need very high precision to measure small changes – this is challenging

• If the half-life is short, the parent decays completely on geological time-scales

Chemistry

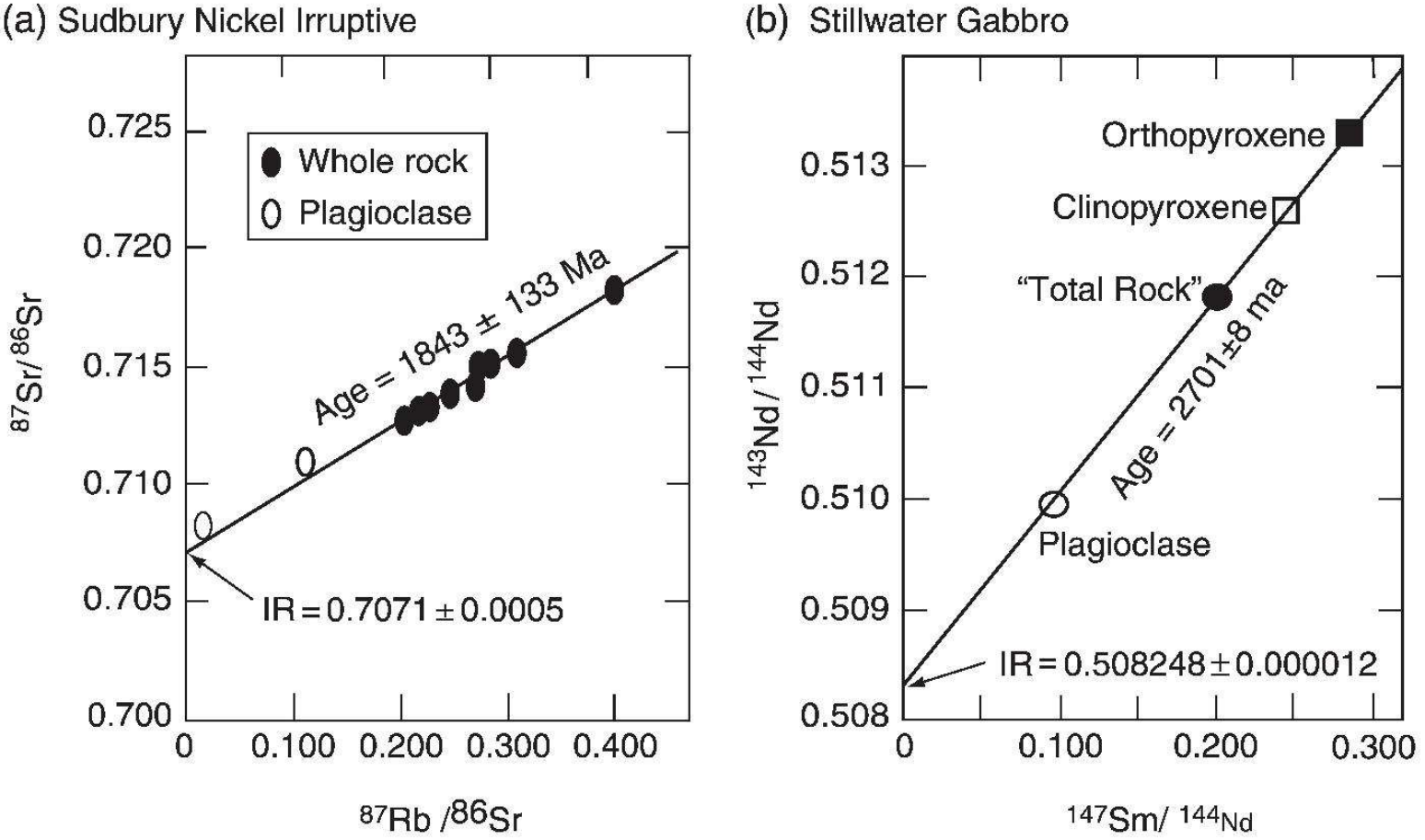
• Parent and daughter have different chemistry.

• Parent and daughter are fractionated among different minerals in the rock,
•i.e. they have different partition coefficients in different minerals.

• Daughter isotope is in low abundance naturally so that the amount of radiogenic isotope is measurable.

Geochronology: Isochrons

Isochron diagrams are graphs used to determine the age of a rock.
A straight line on the diagram is termed an **isochron** (line of equal time).



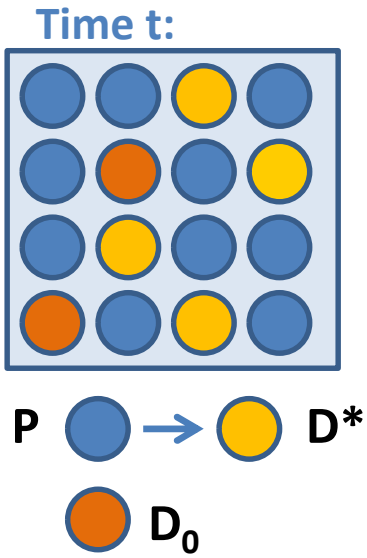
Rb-Sr isochron

Sm-Nd isochron

Misra, Figure 10.5

We have already derived the geochronology equation:

$$D = D_0 + N\,(e^{\lambda t} - 1)$$



N: Number of parent atoms present at time t
D: Number of daughter atoms present at time t
D_0: Number of daughter atoms present at time t = 0
λ: Decay constant

$D = D_0 + D^*$, where D^* = radiogenic daughter atoms
In the example shown, $N_0 = 14$ and $N = 10$
$D = D_0 + D^* = 2 + 4 = 6$

$$D = D_0 + N (e^{\lambda t} - 1)$$

Example: The Rb-Sr system

^{87}Rb decays to ^{87}Sr.

The total number of ^{87}Sr atoms in a rock that has been a closed system for t years is given by:

$$^{87}Sr = {}^{87}Sr_0 + {}^{87}Rb (e^{\lambda t} - 1)$$

^{87}Sr is the total number of Sr atoms present today
$^{87}Sr_0$ is the number of atoms of ^{87}Sr present when the sample first formed
^{87}Rb is the number of atoms of ^{87}Rb present today
λ is the decay constant

$$^{87}Sr = {}^{87}Sr_0 + {}^{87}Rb\,(e^{\lambda t} - 1)$$

• Precise measurement of absolute isotope concentrations is difficult, so isotope ratios are usually determined in a mass spectrometer.

• A stable isotope not involved in the isotopic decay scheme is used as the ratio-ing isotope. In the Rb-Sr system, ^{86}Sr is used. We divided the whole equation by ^{86}Sr:

$$\left[\frac{^{87}Sr}{^{86}Sr}\right]_m = \left[\frac{^{87}Sr}{^{86}Sr}\right]_0 + \left[\frac{^{87}Rb}{^{86}Sr}\right]_m (e^{\lambda t} - 1)$$



Inductively-coupled plasma mass spectrometer

Subscript "m" means that we can measure this in a mass spectrometer.

Subscript "0" refers to the initial ratio of the two isotopes, when the rock formed.



Thermal ionization mass spectrometer

The equation we have produced is the equation of a straight line ($y = mx + c$)

$$\left[\frac{^{87}Sr}{^{86}Sr}\right]_m = \left[\frac{^{87}Sr}{^{86}Sr}\right]_0 + \left[\frac{^{87}Rb}{^{86}Sr}\right]_m (e^{\lambda t} - 1)$$

Measured with a mass spectrometer: y axis

Intercept on y axis

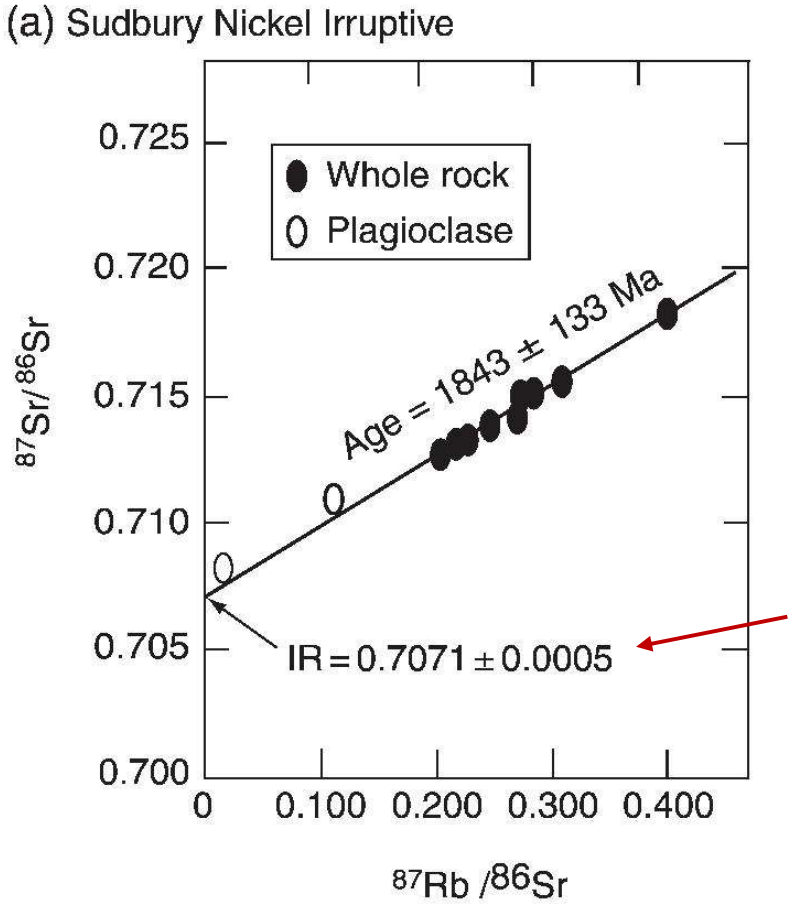
Measured with a mass spectrometer: x axis

Slope of the line on an isochron plot: allows us to determine t

$$y = c + x\,m$$

$$\left[\frac{^{87}Sr}{^{86}Sr}\right]_m = \left[\frac{^{87}Sr}{^{86}Sr}\right]_0 + \left[\frac{^{87}Rb}{^{86}Sr}\right]_m (e^{\lambda t} - 1)$$

y axis | **intercept on y axis** | **x axis** | **slope**

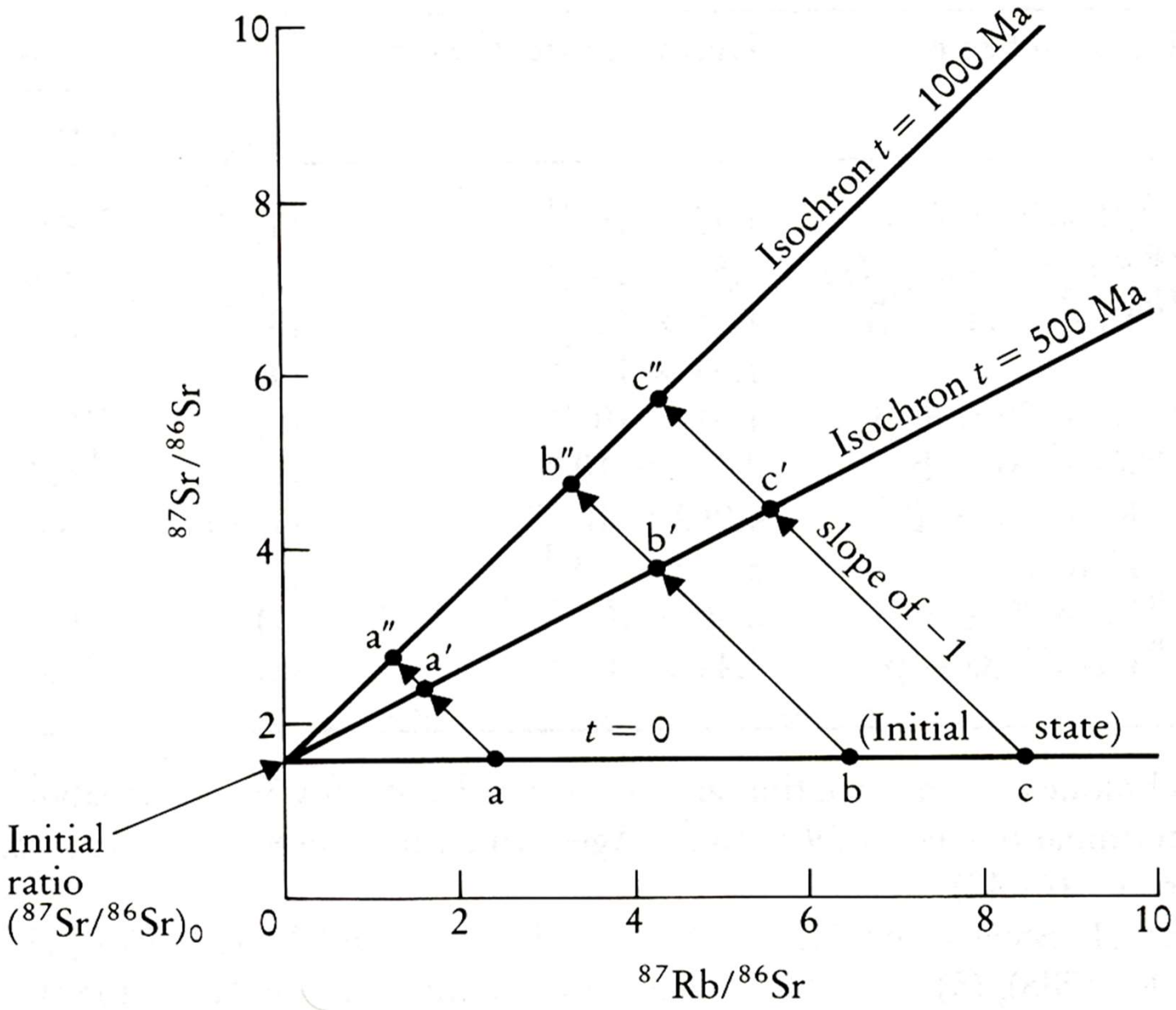


Daughter / Stable

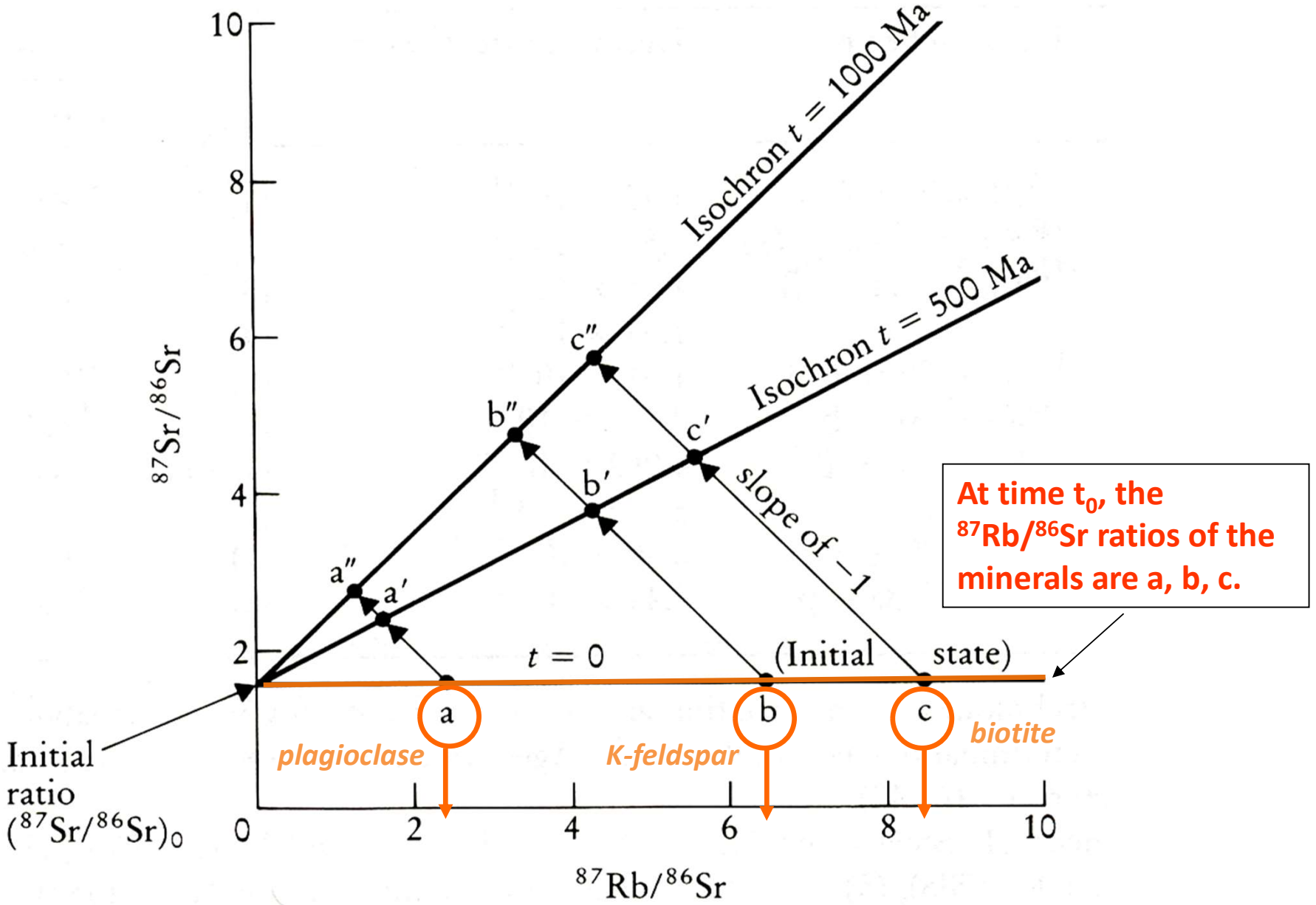
Parent / Stable

Rb-Sr isochron diagram

IR = Initial ratio
i.e. initial $^{87}Sr/^{86}Sr$
or $(^{87}Sr/^{86}Sr)_0$


10
8
6
4
2
0
2
4
6
8
10
$^{87}Sr/^{86}Sr$
$^{87}Rb/^{86}Sr$
Isochron $t = 1000$ Ma
Isochron $t = 500$ Ma
slope of -1
a″
a′
b″
b′
c″
c′
a
b
c
$t = 0$
(Initial state)
Initial ratio $(^{87}Sr/^{86}Sr)_0$

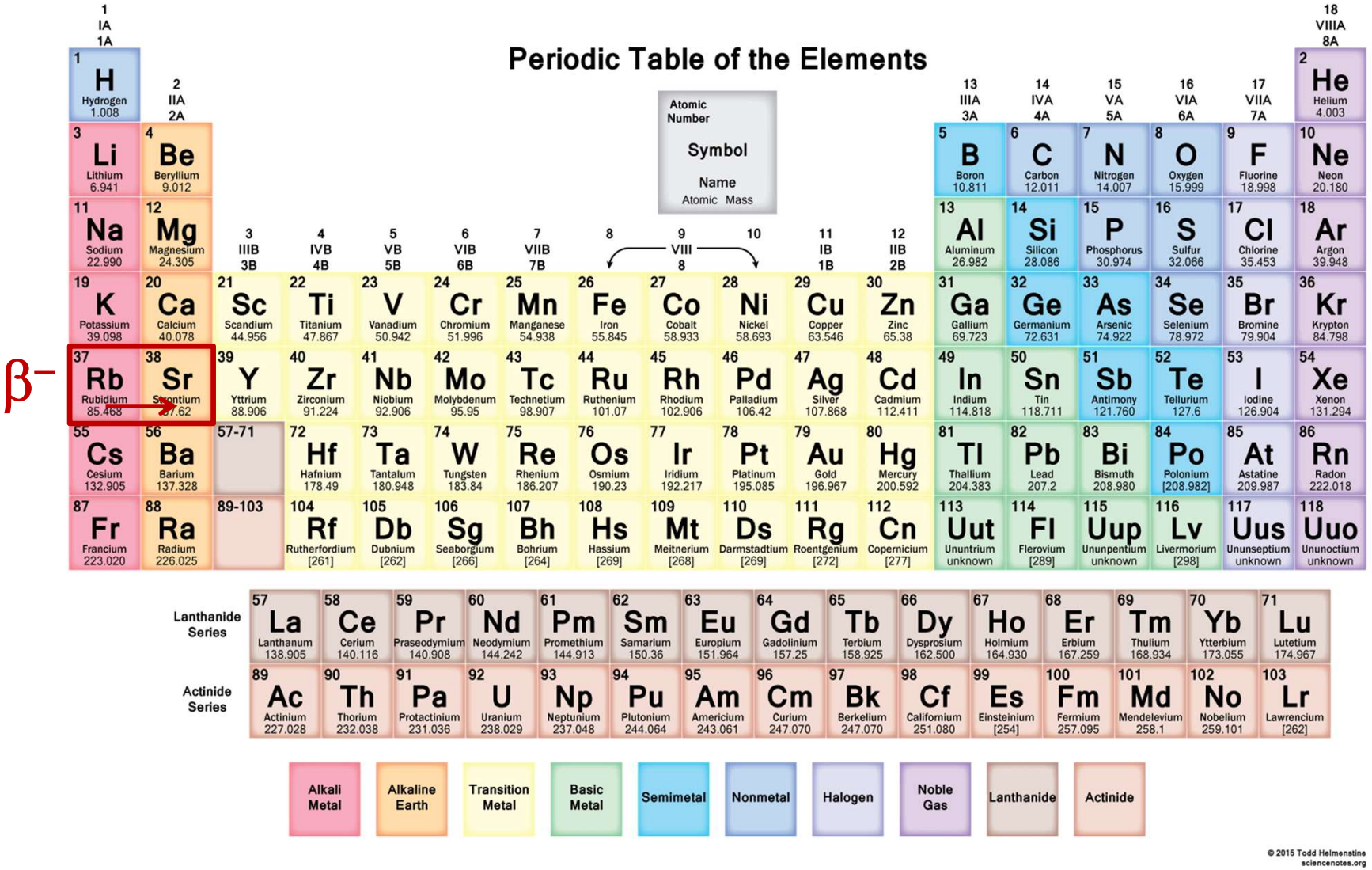
Rollinson



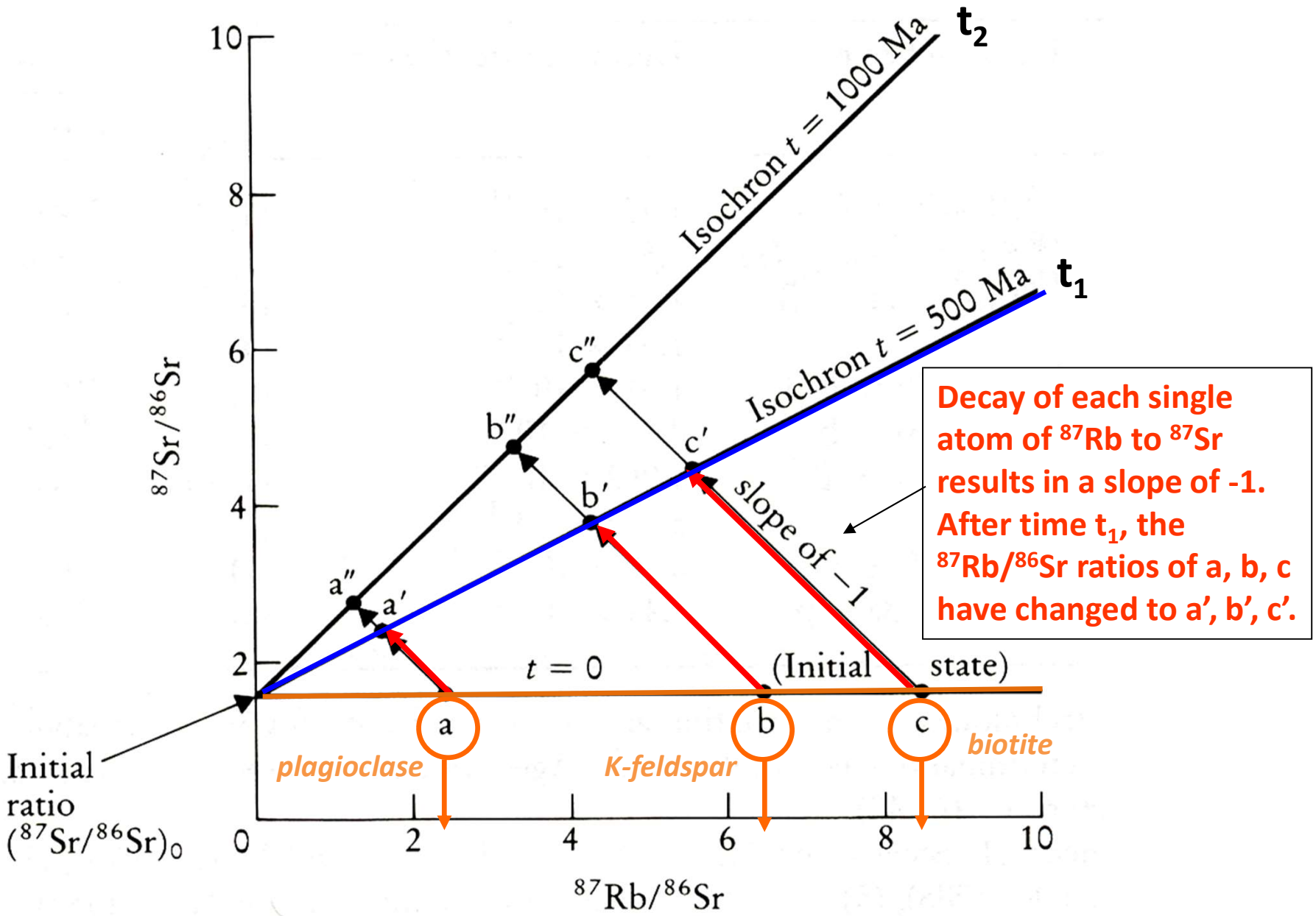
At time t_0, the $^{87}Rb/^{86}Sr$ ratios of the minerals are a, b, c.

Initial $^{87}Rb/^{86}Sr$ ratios for individual minerals in a single rock or individual rocks in a suite

Rollinson

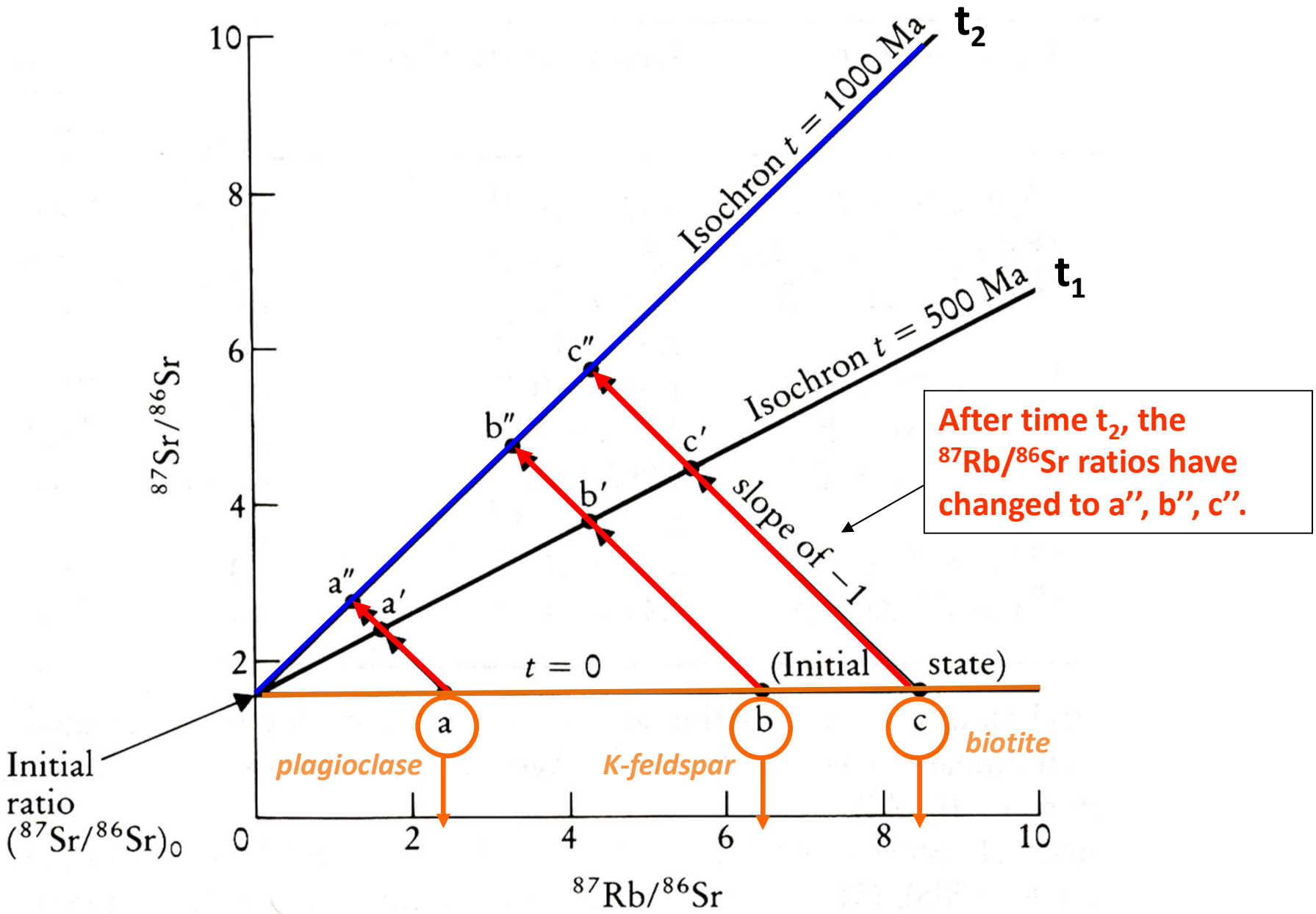


- Rb is similar to K (both are alkali elements, 1+ cations) so K-rich minerals prefer Rb to Sr.
- Sr is similar to Ca (both are alkaline earth elements, 2+ cations), so Ca-rich minerals prefer Sr.

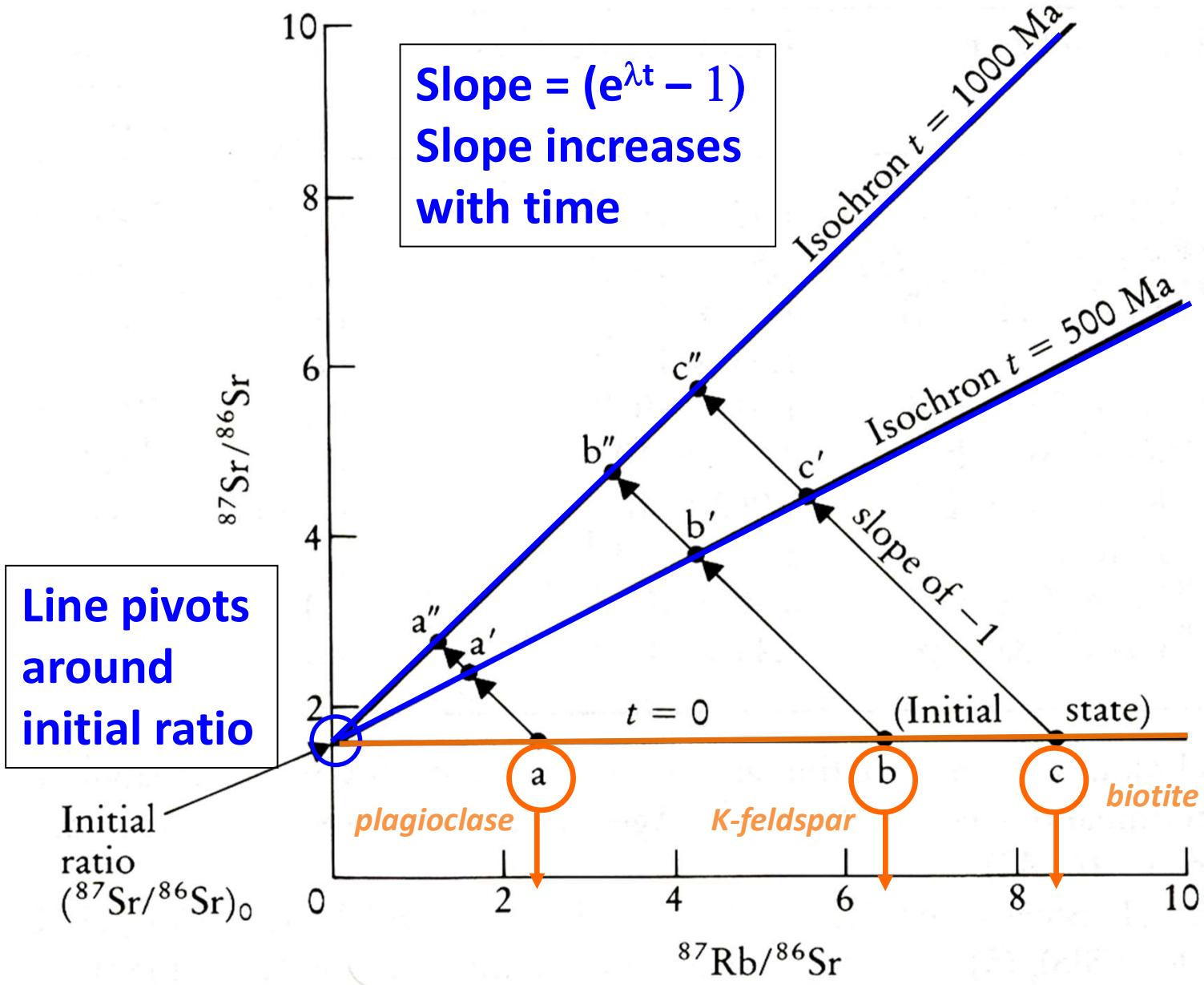


Decay of each single atom of ^{87}Rb to ^{87}Sr results in a slope of -1. After time t_1, the $^{87}Rb/^{86}Sr$ ratios of a, b, c have changed to a', b', c'.

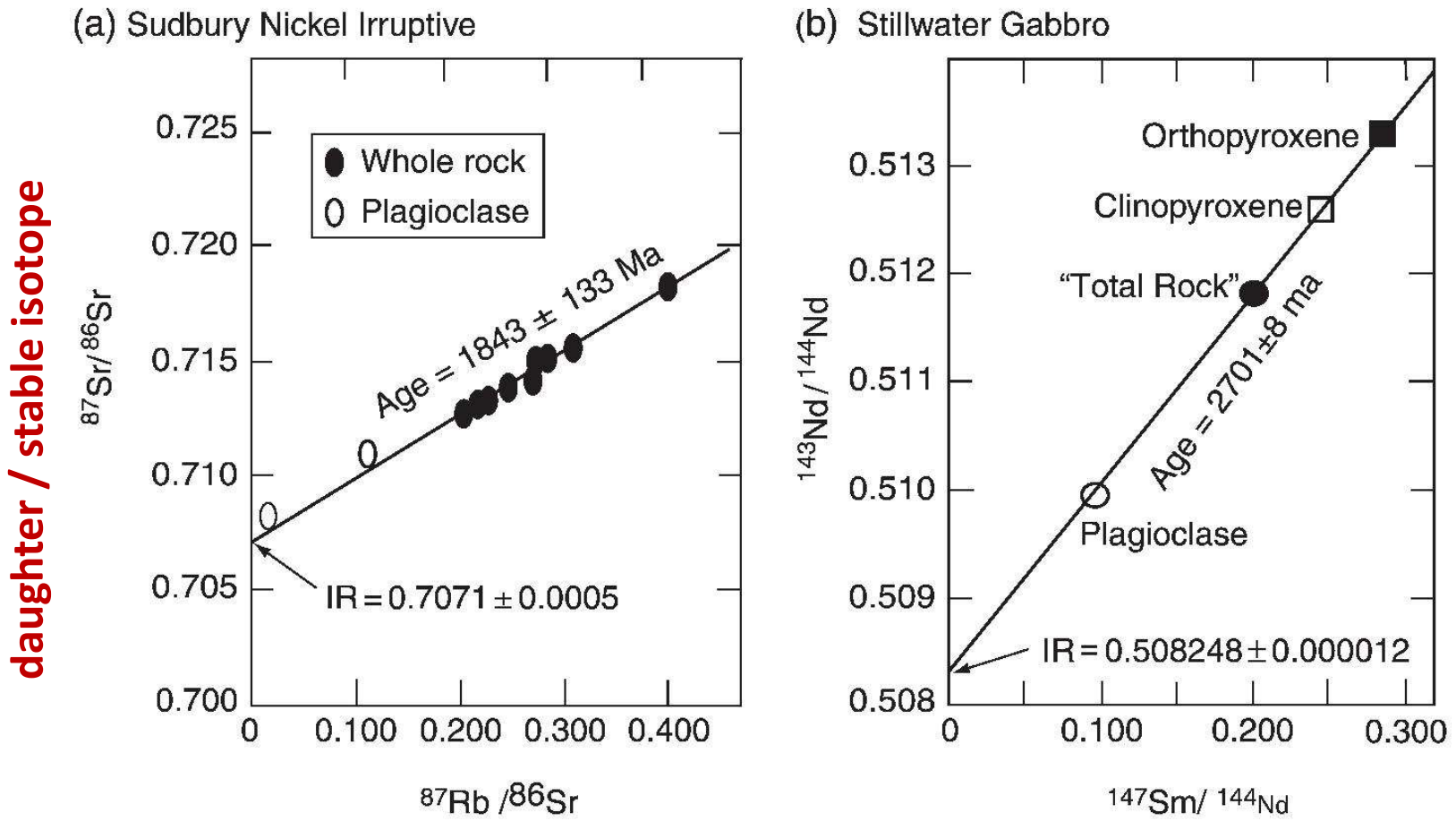
Rollinson


t2
t1
Isochron t = 1000 Ma
Isochron t = 500 Ma
After time t2, the 87Rb/86Sr ratios have changed to a'', b'', c''.
slope of −1
t = 0
(Initial state)
a
b
c
plagioclase
K-feldspar
biotite
Initial ratio (87Sr/86Sr)0
87Sr/86Sr
87Rb/86Sr
0
2
4
6
8
10
a''
a'
b''
b'
c''
c'

Rollinson


Slope = ($e^{\lambda t} - 1$)
Slope increases with time
Line pivots around initial ratio
Isochron t = 1000 Ma
Isochron t = 500 Ma
slope of −1
t = 0
(Initial state)
a
b
c
plagioclase
K-feldspar
biotite
Initial ratio $(^{87}Sr/^{86}Sr)_0$
$^{87}Sr/^{86}Sr$
$^{87}Rb/^{86}Sr$
0
2
4
6
8
10

Rollinson



daughter / stable isotope

parent / stable isotope

$^{87}Rb \rightarrow {}^{87}Sr$

parent / stable isotope

$^{147}Sm \rightarrow {}^{143}Nd$

Misra, Figure 10.5

Chemistry of the Earth: Practical 10

Radioactive isotopes and Geochronology



Practical 7:
Radioisotopes and Geochronology

- **Understand the principles of radioactive decay**
- **Investigate how isochrons are used to determine the ages of rocks**

Basic equation for radioactive decay:

$$N / N_0 = e^{-\lambda t}$$

Basic equation for geochronology:

$$D = D_0 + N (e^{\lambda t} - 1)$$

Example for decay of ^{87}Rb to ^{87}Sr:

$$^{87}Sr = {}^{87}Sr_0 + {}^{87}Rb\,(e^{\lambda t} - 1)$$

Equation for **isochron diagram**:

$$\left[\frac{^{87}Sr}{^{86}Sr}\right]_m = \left[\frac{^{87}Sr}{^{86}Sr}\right]_0 + \left[\frac{^{87}Rb}{^{86}Sr}\right]_m (e^{\lambda t} - 1)$$

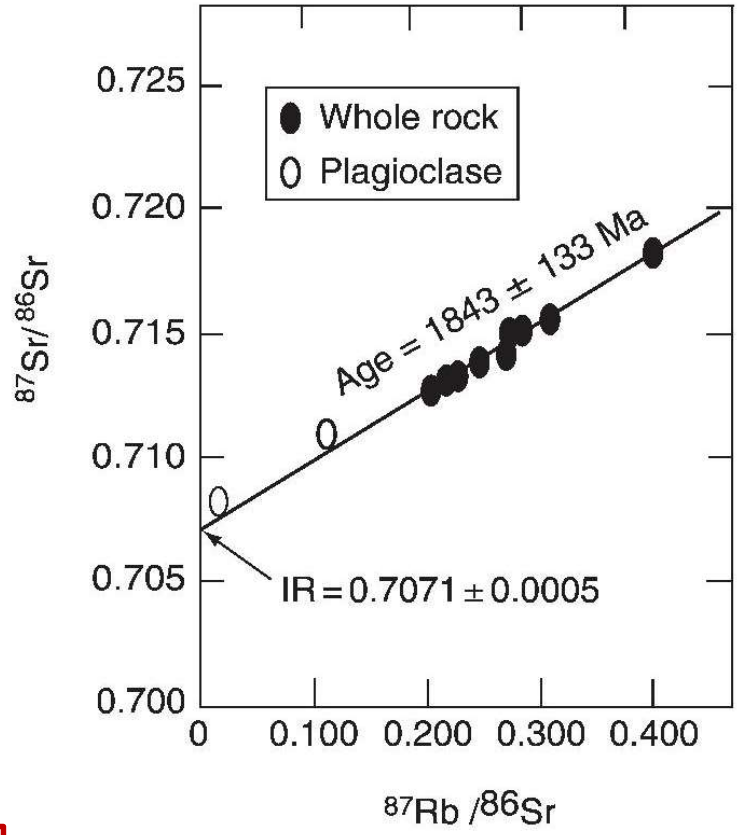
(^{86}Sr is a stable isotope)

λ for ^{87}Rb decay is 1.42×10^{-11} y^{-1}

$t_{½}$ is 4.88×10^{10} y

$$t_{½} = 0.693 / \lambda$$

Isochron diagram for ^{87}Rb-^{87}Sr



Rubidium-Strontium (Rb-Sr) system

Rubidium decay: ^{87}Rb to ^{87}Sr (β^-)

This is a good system for dating intermediate and felsic igneous rocks (e.g. andesite, rhyolite, granite): feldspar-rich rocks.



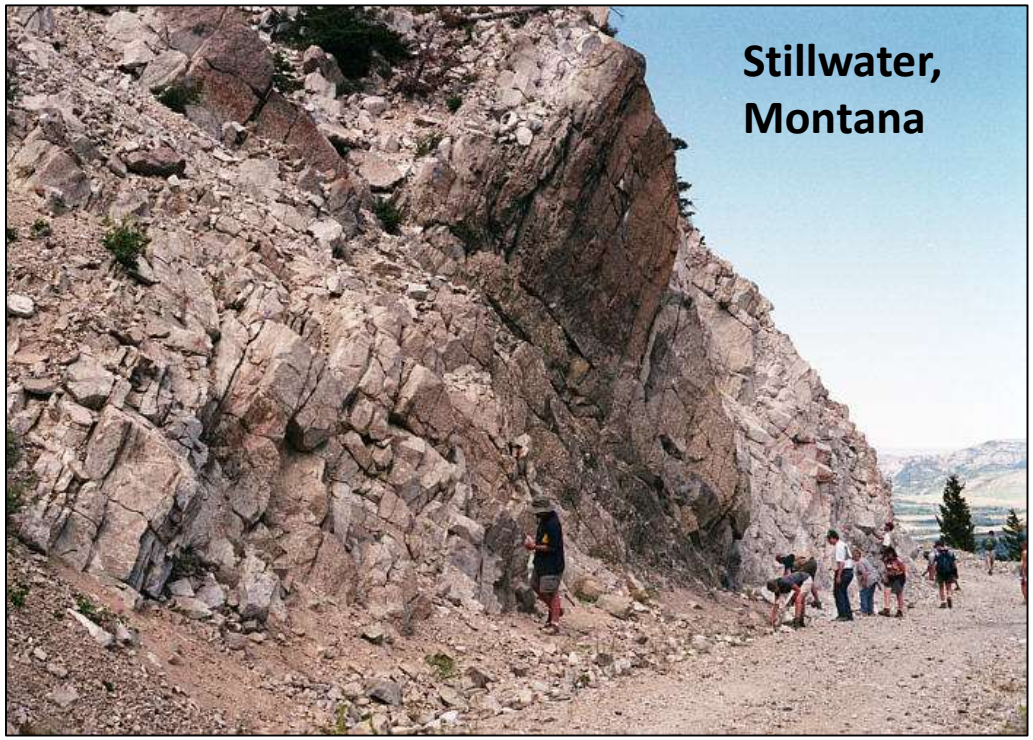
usgs.gov



Samarium – Neodymium (Sm-Nd) system

Samarium decay: ^{147}Sm to ^{143}Nd (α)

This is a good system for dating mafic and ultramafic igneous rocks (e.g. basalt, gabbro): pyroxene-rich rocks.



http://minerva.union.edu/hollochk/teaching_petrology/stillwater.html

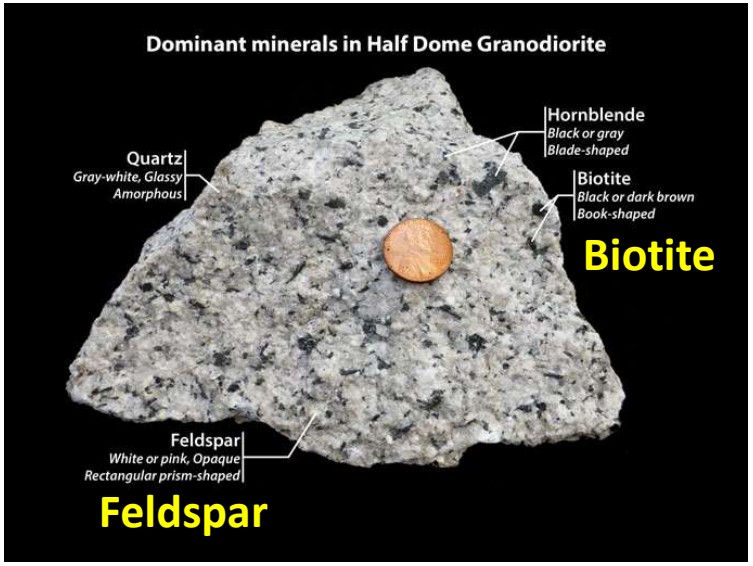


Potassium-Argon (K-Ar) system

Potassium decay:

$^{40}K \rightarrow {}^{40}Ca\ (\beta^-)$ and $^{40}Ar\ (\beta^+$ and e.c.)

Ar is a gas so its presence in a mineral grain is almost entirely radiogenic.

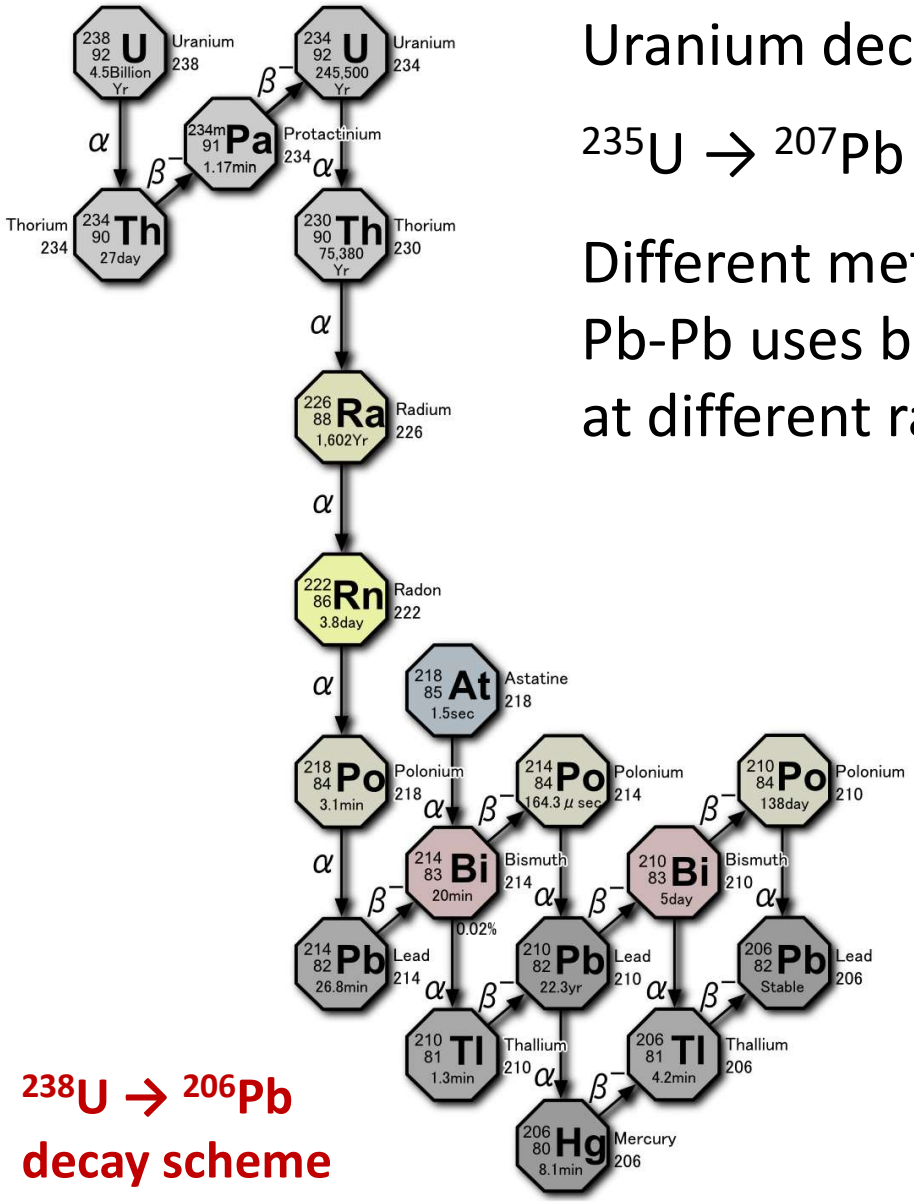


Good for K-bearing minerals e.g. K-feldspar and biotite.

Ar-Ar system

The chronometer is often measured as **^{39}Ar – ^{40}Ar**. This is because it is easier to measure two isotopes of Ar at the same time than to measure K and Ar separately. The sample is irradiated in a nuclear reactor which turns ^{39}K into ^{39}Ar, then ^{39}Ar is measured as a proxy for ^{39}K.

Uranium-Lead (U-Pb) system



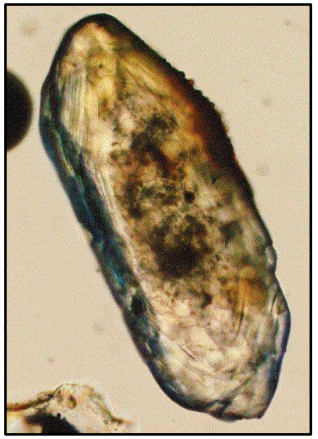
$^{238}U \rightarrow {}^{206}Pb$ decay scheme

Uranium decay:

$^{235}U \rightarrow {}^{207}Pb$ and $^{238}U \rightarrow {}^{206}Pb$

Different methods are used: U-Pb and Pb-Pb. Pb-Pb uses both U-Pb systems, which decay at different rates.

Uranium is often concentrated in zircon, so this is a powerful and precise way to determine the age of zircon.



Zircon grain, 250 μm long.
Wikipedia

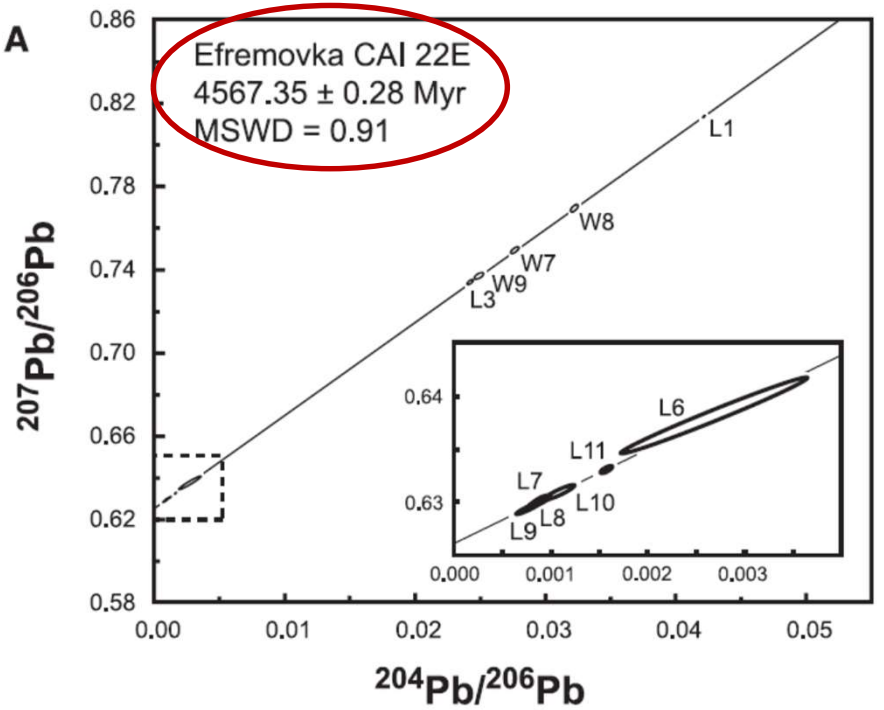
Age of the Solar System: Pb-Pb chronology

Carbonaceous chondrite with a “CAI” inclusion



(Approx 5 cm across)

Connelly et al. (2012) Science, 338, 651-655



W = wash step, L = partial dissolution step, R = residue

Connelly et al (2012): Age of a CAI (number 22E) from the Efremovka carbonaceous chondrite is **4567.35 ± 0.28 Myr**

CAIs give consistent ages of 4567 Ma: these are the oldest objects that have been dated, and we take this to be the age of the Solar System.